



Pool boiling heat transfer of ammonia outside a single tube with fin structures: Hysteresis phenomena and boiling enhancement

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ABSTRACT

Pool boiling heat transfer of ammonia outside a single tube was investigated at three saturation pressures of 6.15, 8.57, and 11.7 bar (saturation temperature of 10, 20, and 30 °C) at the heat fluxes ranging from 1.2 to 72.9 kW/m². Hysteresis in a boiling curve and boiling enhancement were examined on both the smooth tube with machined roughness and two enhanced tubes with low fins and re-entrant cavities. All three tubes have outermost diameters from 15.68 to 15.87 mm and the same length of 796 mm. A large hysteresis between increasing the heat flux and decreasing the heat flux was observed on the smooth tube, and the hysteresis increased as the saturation pressure decreased. The boiling hysteresis of ammonia was reasonably explained from the thermodynamic analysis based on Clausius-Clapeyron and Young-Laplace equations. Heat transfer coefficients on the enhanced tubes were increased by 47% to 105% (Low-fin) and 103% to 159% (Turbo-E) compared with the smooth tube at the saturation pressure of 8.57 bar. However, the hysteresis itself was increased on the enhanced tubes owing to the wall temperature decrease along the fin. Hydrophobic coating and the enhanced tube having re-entrant cavities with cavity mouths of a few micrometers were suggested to mitigate the hysteresis phenomena in the pool boiling of ammonia.

1. Introduction

As global warming is regarded as a serious problem threatening lives and the environment on earth, decarbonization in energy supply and use is becoming more urgent to mitigate its progress. Low-emission fuels such as hydrogen, biofuels, and ammonia are getting more attention as promising candidates for replacing conventional fossil fuels in the electricity, transport, industry, and buildings sectors [1]. Although ammonia has been utilized in various industries including chemical plants and agriculture, the application of ammonia to heating and cooling in industry and building sectors was limited owing to its toxicity and flammability. However, because of its superior thermal capability and zero GWP (Global Warming Potential), ammonia is rising again as the attractive working fluid for absorption or adsorption heat pumps and the efficient carrier for hydrogen ecosystems.

For employing ammonia as a working fluid in heating and cooling applications, it is essential to examine its thermal performance closely in heat-exchanging devices like evaporators and condensers. Pool boiling is a dominant heat transfer phenomenon, especially in a flooded

evaporator which is widely employed in turbo-compressor heat pumps and absorption (adsorption) heat pumps. Because of the long history of its industrial use, there are not a few previous studies about boiling heat transfer of ammonia. Spindler [2] presented a worthwhile overview of pool boiling heat transfer of ammonia concerning various topics including the effects of the number of tubes, surface geometry, and oil contents. Zheng et al. [3] examined the effect of lubricant oil miscible with ammonia on a horizontal plain tube made of carbon steel. They showed that the heat transfer coefficient first decreased with the oil concentration up to 5%, and then increased with a further increase to 10%. A group of Russian scientists [4,5] revealed the effect of surface structures on the pool boiling heat transfer of ammonia by applying the porous coating and mechanical grooves on tube surfaces. They also reported that the fluorocarbon-coated tube exhibited a 3 to 4 times higher heat transfer coefficient compared with the plain tube surface. Inoue et al. [6] investigated boiling performance in binary mixtures of ammonia/water on a horizontal platinum wire with a diameter of 0.3 mm and reported that the boiling heat transfer coefficient in the mixture was considerably smaller than those in both pure fluids. Arima et al. [7] also measured the boiling heat transfer coefficient of the ammonia/water mixture on the

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Nomenclature			
A	area (m^2)	t_w	tube thickness (m)
c_p	specific heat (J/kgK)	T_i	inlet temperature (K)
D	diameter (m)	T_o	outlet temperature (K)
D_f	outmost diameter including fins (m)	T_∞	pool temperature (K)
f	friction factor	U	overall heat transfer coefficient ($\text{W/m}^2\text{K}$)
h_b	boiling heat transfer coefficient ($\text{W/m}^2\text{K}$)	V	voltage (V)
h_i	convective heat transfer coefficient in a pipe ($\text{W/m}^2\text{K}$)	θ	contact angle ($^\circ$)
h_f	fin height (m)	ν_{lv}	specific volume change, $\nu_v - \nu_l$ (m^3/kg)
h_{lv}	heat of vaporization (J/kg)	σ	surface energy (N/m)
L	tube length (m)	ϕ	cone angle of a cavity ($^\circ$)
$\dot{m}$	tube length (m)	ΔT_{lm}	log mean temperature difference (K)
Nu	Nusselt number	Subscripts	
P	pressure (Pa)	D	diameter
Pr	Prandtl number	i	inner
Q	heat transfer rate (W)	l	liquid
q''	heat flux (W/m^2)	o	outer
r_{min}	minimum bubble radius (m)	sat	saturation
R_a	arithmetic average roughness (m)	v	vapor
R_c	cavity radius (m)	w	wall

horizontal silver block (diameter = 10 mm) with a surface roughness of 1 μm . Abbas et al. [8] reviewed previous experimental data about pool boiling heat transfer of ammonia outside a single tube and tube bundles and compared the data with existing correlations. They suggested applying the existing correlations carefully because there is a wide disparity between the studies. Fernández-Seara et al. [9] acquired heat transfer coefficients of both pool boiling and spray evaporation on a horizontal titanium tube with an outer diameter of 19.05 mm, and reported that spray evaporation was superior to pool boiling at the low heat flux of $<20,000 \text{ W/m}^2$, but pool boiling outperformed spray evaporation at the high heat flux and low flow rate. Zeng et al. [10,11] investigated the tube bundle effect on spray evaporation of ammonia with tube bundles having a triangular-pitch pattern and compared the heat transfer coefficient with those of square-pitch tube bundles and pool boiling. Chyu et al. [12] conducted flooded boiling tests of an ammonia/lubricant mixture on an enhanced-tube bundle and showed that the heat transfer coefficient increases from the bottom row to the top row.

There are lots of previous studies about boiling heat transfer improvement on the enhanced tube with fin structures for synthetic refrigerants and hydrocarbons [13–15]. While various boiling enhancement techniques including porous coating, mechanical grooves, and hydrophobic coating [4,5,16] have been introduced for ammonia, there is a limited number of previous studies about the boiling heat transfer of ammonia on a finned surface. Fernández-Seara et al. [17] investigated the heat transfer enhancement of ammonia on tubes with integral fins and obtained a surface enhancement factor of 1.2 on average (20% enhancement) compared with the plain tube. Interestingly, they also reported that there was a hysteresis in a boiling curve of ammonia between increasing the heat flux and decreasing the heat flux, and the hysteresis diminished as time passed. The hysteresis phenomena in a boiling curve are already known in pool boiling using liquids of high wettability such as hydrocarbons and typical refrigerants. Ayub and Bergles [18] compared the hysteresis of boiling curves on four different types of GEWA-T (enhanced) tubes in saturated R-113. Shi et al. [19] analyzed the mechanism of the hysteresis in pool boiling and categorized the hysteresis into two types, (1) hysteresis of temperature overshoot (TOS) at the boiling incipience and (2) hysteresis of temperature deviation (TD) at the development of nucleate boiling. Wojcik [20] investigated the hysteresis of boiling heat transfer on fibrous porous structures and showed that there are three types of hysteresis according

to the liquid wettability and porous layer thickness. Wang et al. [21] also investigated the hysteresis phenomenon in pool boiling of water on the micro-nano biporous structure and demonstrated that PTFE (Polytetrafluoroethylene) coated composite surface exhibited considerable reduction of the hysteresis. The hysteresis phenomena were also noticed in the flow boiling condition. Doretti et al. [22] investigated the hysteresis phenomenon during the flow boiling of R245fa on a carbon/carbon surface by measuring the wall temperature and observing the two-phase flow pattern. Arcasi et al. [23] conducted dedicated flow boiling experiments using R134a in a horizontal stainless steel tube and found out that the heat transfer coefficient increases due to hysteresis when the maximum transient heat flux increases beyond the imposed reference heat flux.

Even if ammonia becomes more important in the heating and cooling sector as a natural refrigerant with zero GWP, the hysteresis of ammonia in pool boiling heat transfer has not yet been clarified not only on the plain surface but on the finned surface. In addition, experimental studies about boiling enhancement of ammonia on the enhanced tubes with fin structures are quite rare. The present study is aimed to investigate the hysteresis phenomena in a boiling curve of ammonia on both smooth and enhanced tubes and the boiling enhancement on the enhanced tubes compared with a smooth one. The mechanism of the hysteresis phenomena is reasonably explained via thermodynamic consideration, and measures to mitigate the hysteresis in pool boiling of ammonia are suggested.

2. Experimental setup

2.1. Experimental apparatus and test tube

Fig. 1 shows schematics of the experimental apparatus and test section for pool boiling of ammonia. A cylindrical test chamber was made of stainless steel (SS 316) and has an inner diameter of 254.4 mm. The chamber wall with a thickness of 6.5 mm was designed to withstand the maximum internal pressure of ammonia (20 bar). The entire length of the chamber including the flanges is 850 mm, and the internal space that is practically wetted with ammonia is 796 mm long. A test tube with an outer diameter of 15.87 mm, which is made of stainless steel, was placed at the bottom center of the test chamber. Teflon housing was inserted between the test tube and the chamber flange to minimize conduction heat loss from the tube to the flange. Inlet and outlet temperatures of the

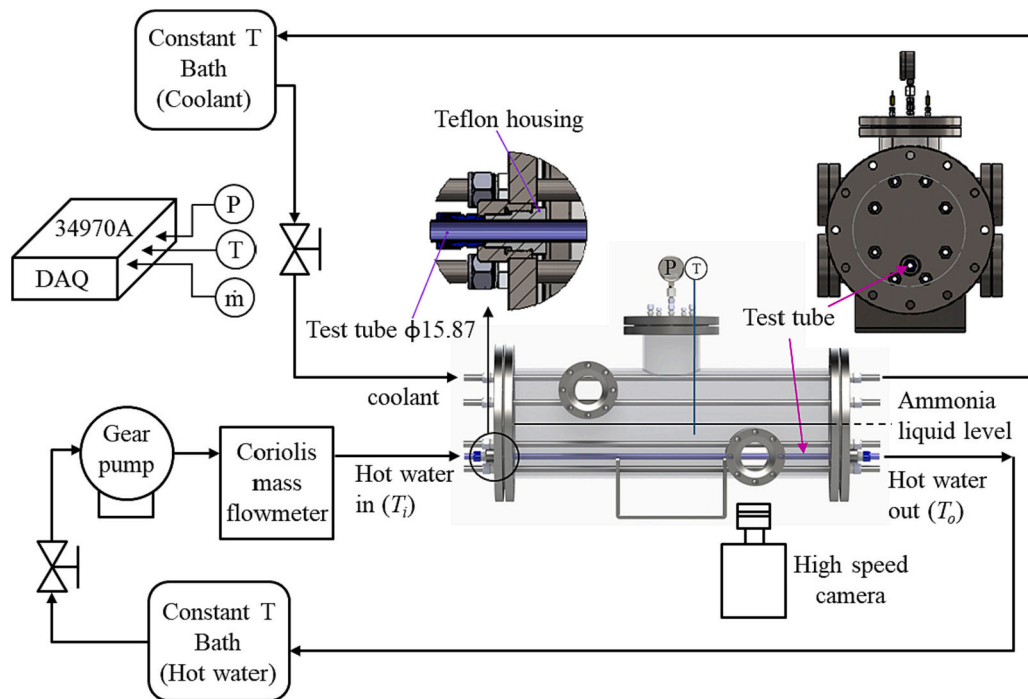


Fig. 1. Schematics of the experimental apparatus and the test section (test chamber and test tube) for pool boiling of ammonia.

test tube were measured using ultra-precise RTD sensors with 1/10 DIN accuracy (OMEGA, P-M-1/10-1/8), which has an accuracy of $\pm 0.03\text{ }^{\circ}\text{C}$ at the temperature of $0\text{ }^{\circ}\text{C}$. The liquid pool temperature of ammonia was measured using T-type thermocouples at two points in the chamber. The internal pressure of ammonia was measured using a pressure transducer connected to the top side of the chamber. The fluid temperature in the test tube was controlled by flowing hot distilled water through the tube, and the temperature of distilled water was maintained by using a constant temperature bath. The water flow rate through the tube was controlled with a gear pump (ISMATEC, MCP-Z) having a volumetric flow rate range from 70 to 7000 ml/min and the exact flow rate was measured with a Coriolis mass flow meter (TRICOR, TCM 0650). The measured mass flow rate of hot water was maintained at 3.8 kg/min throughout the entire test. There are eight cooling tubes embedded in the test chamber, four of which are located in the vapor region (top half), and the other four tubes are placed in the liquid region (bottom half). Four tubes in the vapor region are utilized to condense the ammonia vapor generated from the test tube and the other four tubes in the liquid region are used to maintain the liquid ammonia temperature constant. Every four tubes at the vapor and liquid regions are connected to the recirculating chiller separately. The chamber has two glass windows at the top side (vapor region) and bottom side (liquid region) for visual observation. Bubble behavior on the tube surface was captured using a high-speed camera (Phantom, Miro M320S) with a frame rate of 2000 fps. The specifications of all measurement instruments are summarized in [Table 1](#).

Three tubes with different surface structures were prepared to examine the effect of surface geometry on the boiling phenomena of ammonia. A smooth tube is a plain tube with a machined surface, and two enhanced tubes are low-fin and Turbo-E. The arithmetic average roughness (R_a) of the smooth tube was measured to be 0.296 μm using a surface roughness tester (Mitutoyo, SJ-410). The low-fin tube has many annular fins with a height of 1.33 mm formed by a rolling machine and the outmost diameter of the tube is maintained to be the same as the smooth tube. Turbo-E tube has re-entrant type cavities with a height of 0.76 mm and the outmost diameter is maintained similar to the smooth tube as well. Fig. 2 shows the schematic of the smooth tube and actual images of two enhanced tubes. Inner diameter (D_i), outer diameter (D_o),

Table 1
Specifications of all measurement instruments for the pool boiling test.

Instrument	Model	Measurement range	Accuracy
Hot water in/out Temperature (RTD)	1/10 DIN RTD	−100 to 250 °C	±0.03 °C at 0 °C
Ammonia pool Temperature	T-type Thermocouple	0 to 260 °C	±0.5 °C
Pressure transducer	OMEGA PX409	0 to 250 PSIA	±0.08% reading
Gear pump	ISMATEC MCP-Z	73 to 7241 ml/min	1.2 ml (resolution)
Mass flowmeter	Tricor TCM 650	0 to 650 kg/h	±0.1% reading
Constant T bath (hot water)	MaXircu CL-30	−35 to 150 °C	±0.2 °C (uniformity)
High speed camera	Phantom Miro 320S	2 Megapixels	2000 fps (at 1024 × 1024)

tube thickness (t_w), fin height (h_f), and outermost diameter including fins (D_f) are indicated in Fig. 2(b). Detailed dimensions and materials for three tubes (Smooth, Low-fin, and Turbo-E) are provided in Table 2. The Turbo-E tube was made of carbon steel A179, not stainless steel because stainless steel is too strong to form a cavity structure of Turbo-E.

2.2. Experimental procedure and data reduction

Before filling ammonia into the chamber, the entire chamber is evacuated with a rotary vacuum pump until the internal pressure reaches 3–5 mTorr. Then, liquid ammonia is injected from the supply line at the bottom of the chamber. Once the liquid level of the ammonia pool reaches nearly half of the chamber, the valve of the supply line is closed. The saturated liquid (vapor) temperature of ammonia is controlled to the designated values using eight tubes circumferentially installed in the chamber. In the present study, the boiling phenomena of ammonia were examined at three saturation temperatures of 10, 20, and 30 °C, which corresponds to the saturation pressure of 6.15, 8.57, and 11.7 bar, respectively. Once the liquid pool of ammonia is stabilized at the saturation state, distilled water at a higher temperature than the saturation temperature flows through the test tube. Since the tube wall

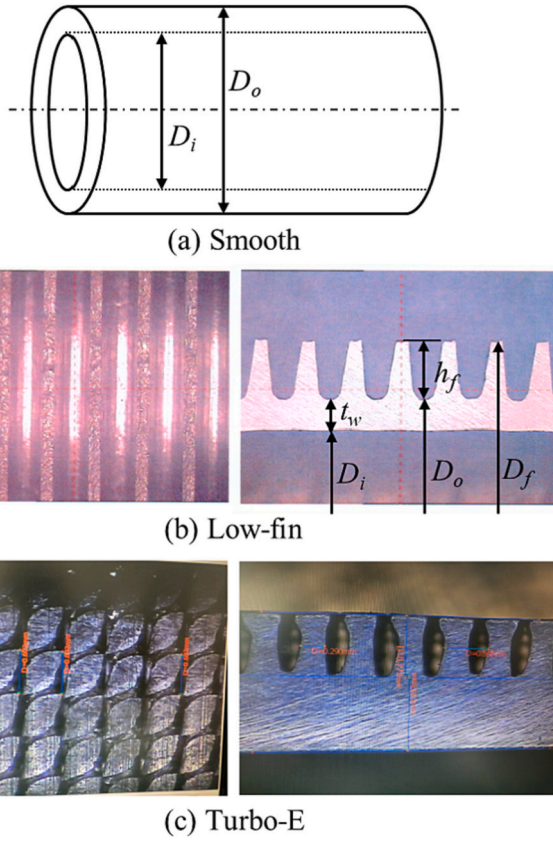


Fig. 2. (a) Schematic drawing of the smooth tube and (b-c) actual images of Low-fin and Turbo-E tubes.

Table 2

Material and detailed dimensions of the smooth tube and two enhanced tubes.

	Material	D_i	D_o	t_w	D_f	h_f	Number of fins (#/inch)
Smooth	Stainless steel 316	13.39	15.87	1.24	–	–	–
Low-fin	Stainless steel 439	11.81	13.07	0.63	15.73	1.33	26
Turbo-E	Carbon steel A179	12.24	14.16	0.96	15.68	0.76	44

temperature is higher than the liquid pool temperature, natural convection or nucleate boiling heat transfer occurs on the tube surface according to the amount of wall superheats. After the tube wall temperature reaches a steady state, all temperature and pressure data are stored for 10 min, and then the inlet water temperature in the tube is increased to the next level. Input heat fluxes in the present study range from 1.2 to 72.9 kW/m², which show a little variation according to the saturation pressure and surface structure. Once the heat flux reaches the maximum value, the increasing heat flux case is terminated and now water temperature through the tube is decreased (decreasing heat flux case). When the water temperature decreases to the initial value, the entire test is terminated.

Heat transfer rate, Q was calculated from the enthalpy change of hot water through the test tube as shown in Eq. (1).

$$Q = \dot{m}_p(T_i - T_o) \quad (1)$$

where T_i and T_o refer to the inlet and outlet water temperature of the test tube, respectively. The total heat transfer rate can also be obtained by multiplying the overall heat transfer coefficient, U , heat transfer area,

A , and log mean temperature difference (LMTD), ΔT_{lm} .

$$Q = UA\Delta T_{lm} \quad (2)$$

$$\Delta T_{lm} = \frac{\Delta T_i - \Delta T_o}{\ln(\Delta T_i / \Delta T_o)} \quad (3)$$

where $\Delta T_i = T_i - T_\infty$, $\Delta T_o = T_o - T_\infty$.

T_∞ is the ammonia pool temperature surrounding the test tube. Heat is transferred from hot distilled water to the ammonia pool by internal convection inside the tube, conduction through the tube wall, and boiling outside the tube. This one-dimensional heat transport is expressed in the following thermal resistance network model (Eq. (4)).

$$\frac{1}{UA} = \frac{1}{h_i A_i} + \frac{\ln(D_o/D_i)}{2\pi k L} + \frac{1}{h_b A_o} \quad (4)$$

where h_i indicates the internal convective heat transfer coefficient and A_i refers to the inner surface area of the tube. L is the wetted length of the test tube, and A_o is the outer surface area of the tube calculated in the following Eq. (5).

$$A_o = \pi D_o L \quad (5)$$

The definition of the outer surface area, A_o , is consistent for both smooth and enhanced tubes. That is, A_o is calculated based on the outer diameter, D_o , designated in Fig. 2 and summarized in Table 2. The low fins (Low-fin tube) or cavities (Turbo-E tube) are considered additional features similar to the surface coating for enhancing boiling heat transfer. The convective heat transfer coefficient of the internal flow can be obtained from the previous semi-empirical correlations. Since the Reynolds number of the internal flow ranges from 5266 to 14,733, it is appropriate to select Gnielinski's correlation (Eq. (6)) which is valid in the Reynolds number range from 3000 to 5×10^6 [24].

$$Nu_D = \frac{(f/8)(Re_D - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \quad (6)$$

where $f = (0.790 \ln Re_D - 1.64)^{-2}$

Convective heat transfer coefficient (h_i) is acquired from Eq. (6) and the boiling heat transfer coefficient outside the tube (h_b) can be obtained by substituting the h_i into Eq. (4). Lastly, wall superheat, ΔT_w can be calculated from the heat transfer rate, Q , and the heat transfer coefficient, h_b .

Measurement uncertainties of the important variables were analyzed according to the guidelines of the relevant reference [25]. The ultra-precise RTD sensor (1/10 DIN class) has ± 0.03 °C tolerance at the measuring temperature of 0 °C, and T-type thermocouples with special limits of the error have an accuracy of ± 0.5 °C. The mass flow rate was acquired with an uncertainty of $\pm 0.1\%$ of the reading. As a result, averaged expanded uncertainties of heat transfer rate (Q), boiling heat transfer coefficient (h_b), and wall superheat (ΔT) were calculated to be 4.2%, 18.0%, and 19.0%, respectively.

3. Results and discussion

3.1. Hysteresis in a boiling curve on the smooth tube

Fig. 3 shows the boiling curve on the smooth surface at the saturation pressure of 8.57 bar and corresponding saturation temperature of 20 °C. It is noticeable that the boiling curve exhibits large hysteresis between when the heat flux increases and when the heat flux decreases. In the case of increasing the heat flux, a larger wall superheat (ΔT_w) is observed at the low heat flux region, which indicates lower boiling heat transfer performance owing to the suppression of the bubble nucleation. In contrast, lower wall superheat is obtained in case of decreasing the heat flux, implying higher boiling heat transfer performance. Both boiling curves of increasing and decreasing the heat flux show similar wall

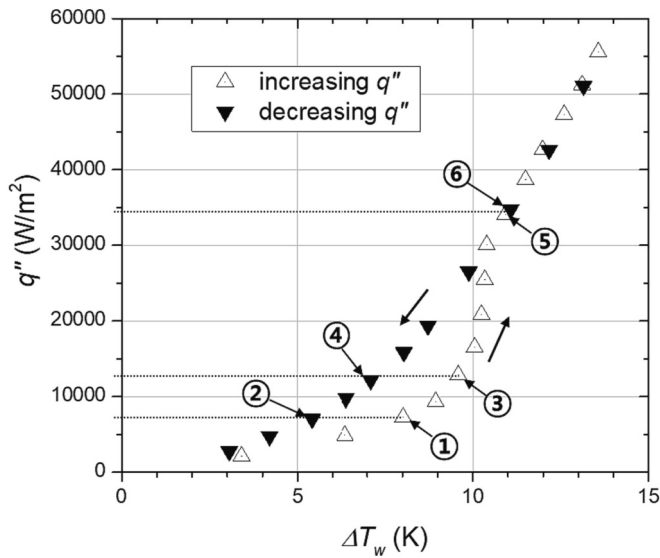


Fig. 3. Hysteresis in boiling curves on the smooth surface at the saturation pressure of 8.57 bar ($T_{sat} = 20\text{ }^{\circ}\text{C}$).

superheats at the heat flux of approximately $30,000\text{ W/m}^2$. At the higher heat flux region above $30,000\text{ W/m}^2$, both curves are nearly the same regardless of whether the heat flux increases or decreases. As presented in Fig. 4, the photographs taken at three heat fluxes of about 7000, 12,000 and $34,000\text{ W/m}^2$ clearly show the different boiling behaviors when increasing the heat flux and decreasing the heat flux. At the low heat flux of 7254 W/m^2 when increasing the heat flux, no bubble nucleation is observed except one nucleation site throughout the entire tube surface in the picture. In contrast, at a similar heat flux of 7039 W/m^2 when decreasing the heat flux, many bubbles are generated all over the smooth tube surface. When the heat flux increases to $12,859\text{ W/m}^2$ (increasing q''), more bubbles generate on the tube surface, but the number of nucleation sites falls much short of that of the case of decreasing the heat flux. Finally, when the heat flux increases up to $34,100\text{ W/m}^2$, bubble nucleation appears comparable to the case of decreasing the heat flux. These hysteresis phenomena in the boiling curve are sometimes observed in highly wetting fluids. Concerning ammonia, Fernández-Seara et al. [17] reported large hysteresis in the boiling curve between when the heat flux increases and when the heat flux decreases. They also reported that bubble nucleation in case of increasing the heat flux was augmented after continuous heat input for 10 h.

The hysteresis in a boiling curve of ammonia is highly related to the wetting characteristic of liquid ammonia on a solid surface. The surface energy of ammonia at the temperature of $20\text{ }^{\circ}\text{C}$ is 21.7 mN/m , which is relatively small compared with the surface energy of water, 72.7 mN/m ,

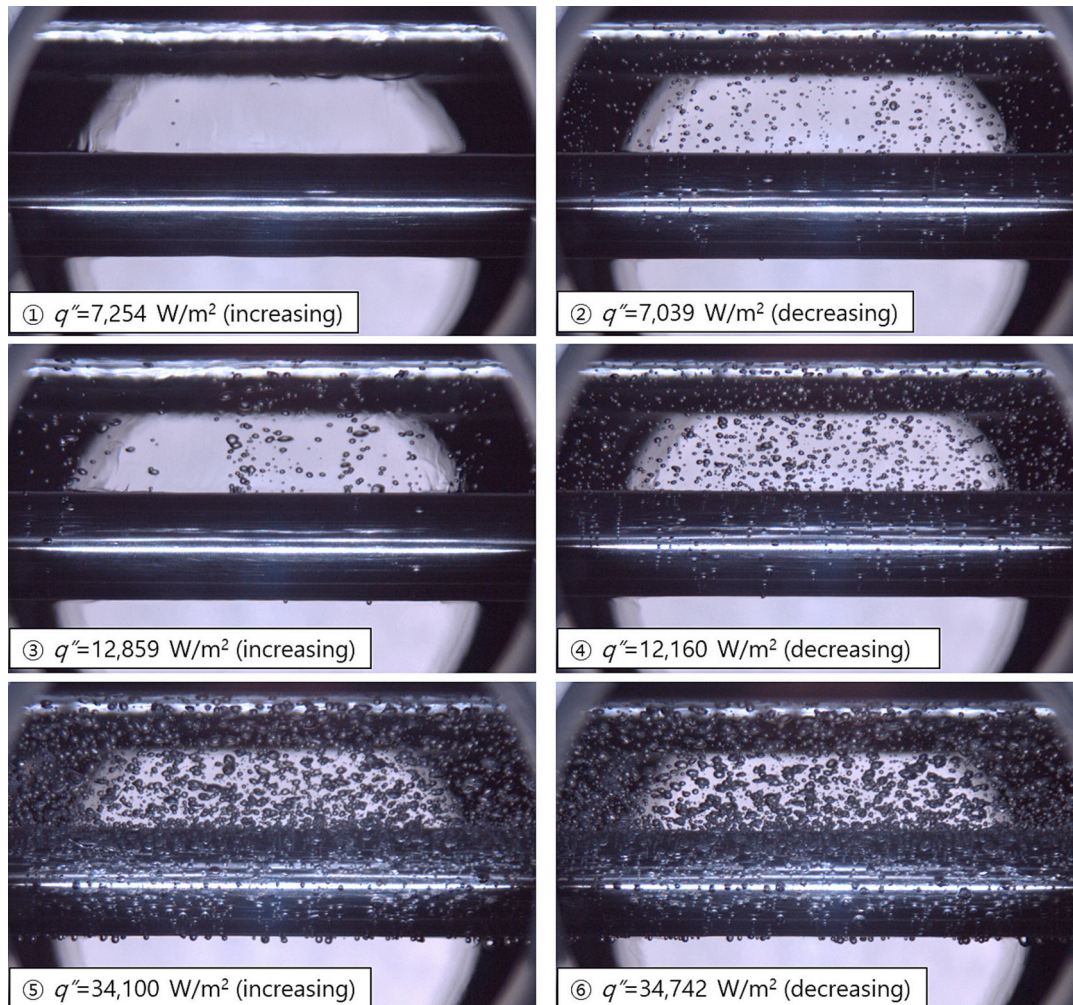


Fig. 4. Captured images of bubble nucleation at three heat fluxes of about 7000, 12,000, and $34,000\text{ W/m}^2$ when increasing the heat flux and decreasing the heat flux.

at the same temperature. The lower surface energy results in a lower equilibrium contact angle with the stainless steel surface. Based on the Clausius-Clapeyron equation and Young-Laplace equation, the wall superheat required to generate bubble nucleation is expressed in the following Eq. (7).

$$T_l - T_{sat}(P_l) > \frac{2\sigma T_{sat}(P_l)v_{lv}}{h_{lv}r_{min}} \quad (7)$$

where T_l is the temperature of the superheated liquid layer, which is approximately the same as the superheated wall temperature, T_w , therefore the left-hand side of Eq. (7) is identical to the wall superheat, ΔT_w . v_{lv} indicates the specific volume change, $v_v - v_l$, and r_{min} refers to the minimum radius of the embryonic bubble. It is inferred from Eq. (7) that a large wall superheat is required to grow the embryonic bubble when the minimum bubble radius is small. The problem is that the minimum radius of the bubble is highly related to whether the heat flux is increasing or decreasing. For highly wetting fluids like ammonia, when the heat flux is increasing the minimum bubble radius occurs at the initial stage of the embryonic bubble inside the cavity as depicted in Fig. 5(a). The small embryonic bubble requires a large amount of wall superheat to grow beyond the wall cavity. In comparison, when the heat flux is decreasing, the bubble growth starts from the remaining bubble after the departure of the previous bubble (Fig. 5(b)). Since the remaining bubble is much larger than the embryonic bubble in the cavity, it is derived from Eq. (7) that a small amount of wall superheat is sufficient to grow the bubble. The cavity radius, R_c is considered in the same order of magnitude as the arithmetic average roughness of the tube ($R_a \sim 0.296 \mu\text{m}$). If we assume that the minimum radius of the embryonic bubble is half of the cavity radius and the minimum radius of the remaining bubble after the departure is twice the cavity radius, the required wall superheats for bubble growth are calculated to be 10.7 K and 2.7 K, respectively, at the saturation temperature of 20 °C. It is concluded that the minimum bubble radius, r_{min} is the most important parameter to decide how large the hysteresis is. Suggestions to minimize the hysteresis in pool boiling of highly wetting fluids are presented in the last section.

The hysteresis phenomena in the boiling curve are highly affected by the saturation pressure. Fig. 6 shows boiling curves of ammonia at three saturation pressures of 6.15, 8.57, and 11.7 bar, when the heat flux increases and when the heat flux decreases. It is noted that the hysteresis between increasing the heat flux and decreasing the heat flux is more pronounced as the saturation pressure decreases. At the saturation pressure of 6.15 bar, the maximum wall superheat before the nucleate boiling reaches 13.9 K in case of increasing the heat flux, and the difference of wall superheats between when increasing the heat flux and when decreasing the heat flux is as high as 3.3 K. As the saturation pressure increases ($P_{sat} = 11.7$ bar), the maximum wall superheat before the nucleate boiling decreases to 8.9 K, and the deviation between two boiling curves is also reduced to 2.2 K. Required wall superheats calculated from Eq. (7) are summarized in Table 3 according to the

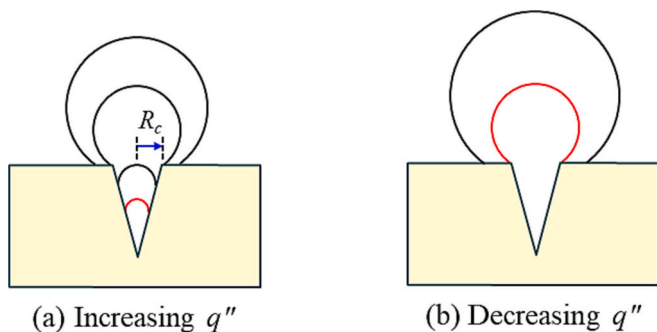


Fig. 5. Comparison of bubble growth in the cavity between when (a) increasing the heat flux and (b) decreasing the heat flux.

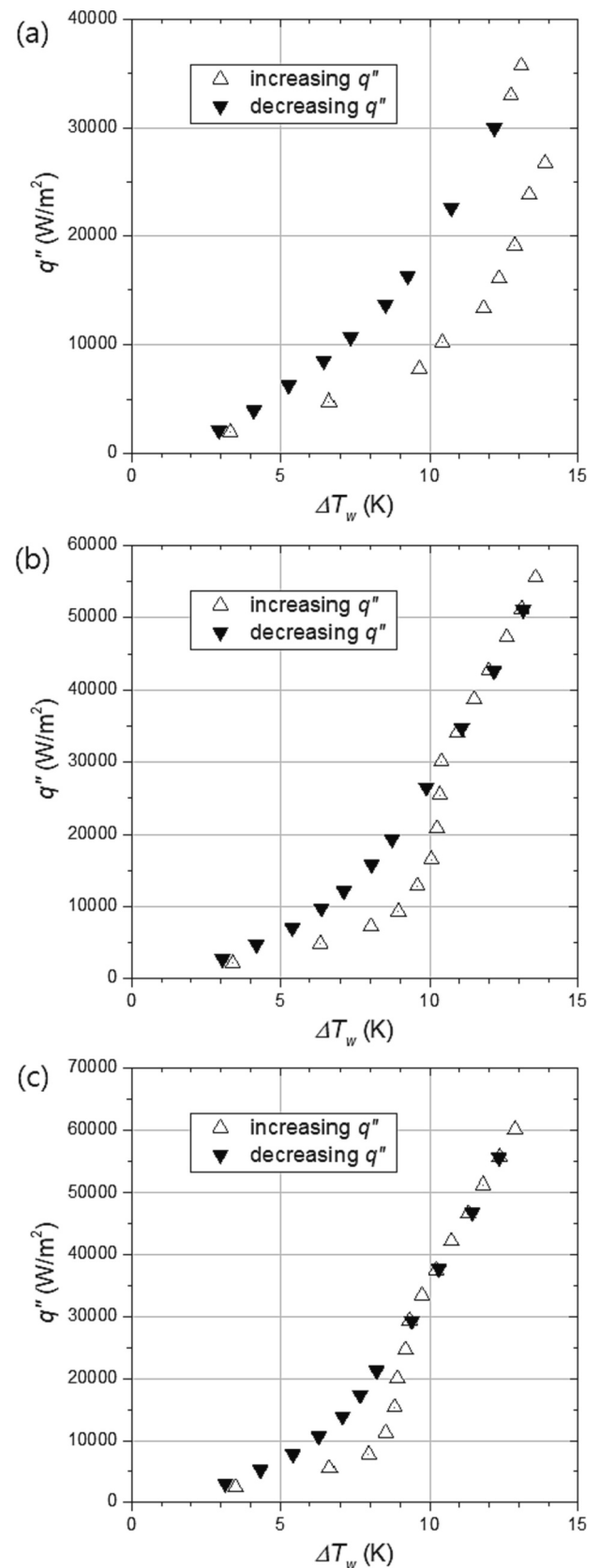


Fig. 6. Boiling curves of ammonia at three saturation pressures of 6.15, 8.57, and 11.7 bar.

Table 3

Wall superheats required for bubble growth according to the saturation pressure.

P_{sat} (bar)	T_{sat} (°C)	Required ΔT (K)	
		increasing ($r_{min} = R_c/2$)	Decreasing ($r_{min} = 2R_c$)
4.29	0	22.2	5.5
6.15	10	15.3	3.8
8.57	20	10.7	2.7
11.7	30	7.5	1.9

saturation pressure. The wall superheat required for the bubble growth at the saturation pressure of 6.15 bar is as much as 15.3 K in case of increasing the heat flux, which is comparable to the maximum wall superheat in the boiling curve in Fig. 6(a). Likewise, required wall superheats at higher saturation pressures of 8.57 and 11.7 bar decrease to 10.7 and 7.5 K, respectively, which are comparable to the maximum wall superheats in the boiling curves in Fig. 6(b-c). As both surface tension (σ) and specific volume change (v_{vl}) decrease with increasing the saturation pressure, it is reasonable that the required wall superheat decreases with the pressure from Eq. (7).

3.2. Hysteresis and boiling enhancement on the enhanced tubes

Boiling curves on enhanced tubes with extended structures of low fins (Low-fin) and re-entrant cavities (Turbo-E) are compared with that on the smooth tube. Fig. 7 shows the boiling curves on the Low-fin tube at the saturation pressure of 8.57 bar ($T_{sat} = 20^\circ\text{C}$) when increasing the heat flux and when decreasing the heat flux, along with boiling curves on the smooth tube. Boiling curves on the Low-fin tube move to the left in both cases of increasing and decreasing the heat flux, implying boiling heat transfer performance is enhanced. At the same time, it is also clear that the hysteresis in boiling curves is enlarged on the Low-fin tube, that is, a larger discrepancy between boiling curves when increasing the heat flux and when decreasing the heat flux is observed. Boiling heat transfer performance on the Low-fin tube is augmented because the heat transfer area increases on the Low-fin tube both when increasing and decreasing the heat flux. On the contrary, the number of bubble nucleation sites increases only in case of decreasing the heat flux, not in case of increasing the heat flux. The difference in bubble nucleation on the finned surface is related to the temperature gradient along the fin on the

Low-fin tube. The wall superheat in the boiling curve on the Low-fin tube (Fig. 7) is based on the wall temperature at the fin base. Because there is conduction resistance in the low fin, the wall temperature at the fin tip is lower than that on the fin base. As we already analyzed from Eq. (7), a large amount of wall superheat is required to grow the bubble when the heat flux is increasing. As a result, the number of nucleation sites is small when increasing the heat flux because the wall superheat at the fin surface is not large enough to generate the bubble. In comparison, more bubble nucleation occurs on the Low-fin tube when the heat flux decreases because the wall superheat at the fin surface is sufficient to grow the bubble, despite the wall temperature decrease along the fin.

The enlarged hysteresis in boiling curves on the Low-fin tube is clarified from the bubble nucleation images presented in Fig. 8. Four images (Fig. 8(a-d)) were taken at similar heat fluxes of approximately $10,000 \text{ W/m}^2$ along the dashed line marked in Fig. 7. When the heat flux increases (Fig. 8(a,c)), number of nucleation sites is small for both smooth and Low-fin tubes. Bubbles are generated at four or five sites on the smooth tube, in contrast, bubbles are only generated at one or two sites on the Low-fin tube, and a couple of bubbles seem agglomerated at the single site. Bubble nucleation seems more suppressed on the Low-fin tube when increasing the heat flux. On the contrary, when the heat flux decreases (Fig. 8(b,d)), the number of nucleation sites is enormously increased on the Low-fin tube. Although bubble generation increases a lot on the smooth tube as well, more bubbles are generated on the Low-fin tube compared with the smooth tube. It is seen from the captured images that larger hysteresis is expected in boiling curves on the Low-fin tube. Along the dashed line in Fig. 7, the wall superheat difference between points (c) and (d) on the Low-fin tube is as much as 4.5 K, compared with 2.6 K between points (a) and (b) on the smooth tube.

The Turbo-E tube has re-entrant cavities on the surface, which may facilitate nucleate boiling heat transfer considerably. Boiling curves on the Turbo-E tube move to the left a lot compared with those on the smooth tube as depicted in Fig. 9. Though boiling performance on the Turbo-E tube is much enhanced, it still has quite a large hysteresis at the low heat flux region. The hysteresis on the Turbo-E tube originates from the same reason as that on the Low-fin tube. There is a temperature gradient along the fin on the Turbo-E tube, which results in the decrease of wall superheat as getting close to the fin tip. The decrease in wall superheat along the fin delays the growth of the bubble on the fin surface when the heat flux increases. Figs. 8(e-f) compare different bubble nucleation behavior according to whether the heat flux increases or decreases, at the heat flux near $10,000 \text{ W/m}^2$. A much larger number of bubble nucleation is observed over the entire tube surface when the heat flux decreases (Fig. 8(f)), while bubble generation is only confined to the small part of the tube when the heat flux increases (Fig. 8(e)). One more interesting point is that the bubble departure diameter on the Turbo-E tube is tremendously increased compared with those on the smooth tube or Low-fin tube. Since the departure diameters on the Turbo-E tube appear to be twice or three times larger than those on the Low-fin tube, the volume of the departed bubbles can be 8 to 27 times larger. This can bring about a huge difference in heat transfer performance because the bubble volume is directly related to the amount of latent heat during the nucleate boiling. The larger bubble departure diameter on the Turbo-E tube is due to bubble nucleation occurring inside the cavity of the tube. The bubble nucleation process on the Turbo-E tube is presented in Fig. 10, in comparison with that on the smooth tube or Low-fin tube. In the case of the smooth tube, the growing bubble departed from the wall when the buoyant force of the bubble is large enough to overcome the surface tension. In the case of the Turbo-E tube, at first, small vapor embryos begin to grow at the bottom region inside the re-entrant cavity because the surface temperature is the highest at the fin base. The growing bubble departed from the fin base region inside the cavity when it is large enough to overcome the surface tension. Since there are a couple of active nucleation sites inside the cavity, the departed bubbles from adjacent sites agglomerate before passing through the cavity mouth. Therefore, the final bubble escaping from the cavity on the

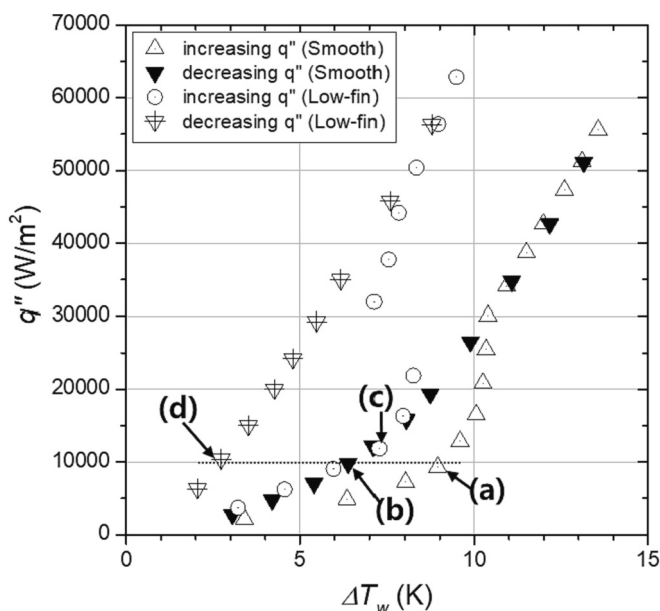


Fig. 7. Boiling curves on the Low-fin tube compared with the smooth tube at the saturation pressure of 8.57 bar ($T_{sat} = 20^\circ\text{C}$).

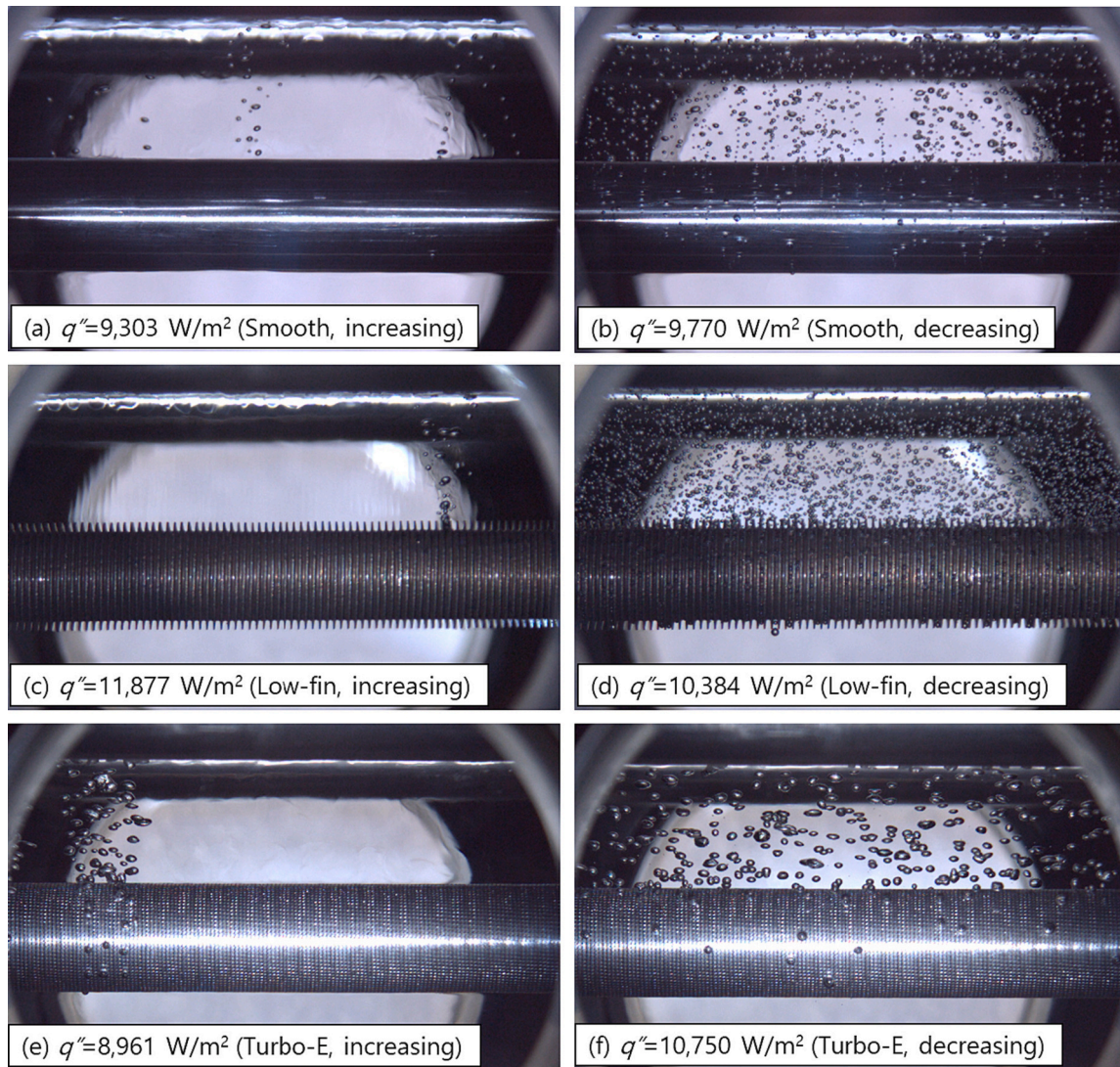


Fig. 8. Comparison of bubble nucleation on the smooth tube and two enhanced tubes (Low-fin and Turbo-E tubes) at the heat flux of approximately 10,000 W/m² ($P_{sat} = 8.57$ bar).

Turbo-E tube is much larger than the departed bubble on the smooth surface.

In Fig. 11, boiling heat transfer coefficients on three tubes (smooth, Low-fin, and Turbo-E) are compared at the saturation pressure of 8.57 bar ($T_{sat} = 20$ °C). Boiling performance is certainly enhanced on both Low-fin and Turbo-E tubes, regardless of whether the heat flux is increasing or decreasing. However, when the heat flux is increasing, the enhancement of the heat transfer coefficient is relatively small at the low heat flux region below 10,000 W/m². The lower enhancement is ascribed to the delay of bubble nucleation on the finned surfaces owing to the reduction of wall superheat along the fin, as we already discussed above. Nucleate boiling on the Turbo-E tube becomes dominant at the heat flux above 10,000 W/m², while nucleate boiling on the Low-fin tube is still weak until the heat flux reaches 30,000 W/m². The difference between the boiling performances of the two tubes reveals that re-entrant cavities are effective for facilitating the bubble nucleation on the finned surface. The heat transfer coefficients on Low-fin and Turbo-E tubes are augmented by 47% and 103% on average compared with that on the smooth tube when the heat flux increases. When the heat flux decreases, the enhancement of heat transfer coefficients is increased to 105% and 159% on average on Low-fin and Turbo-E tubes compared with that on the smooth tube.

3.3. Measures to mitigate hysteresis and improve boiling

In the preceding sections, it is confirmed that the large hysteresis in a boiling curve of ammonia is highly related to the minimum radius (r_{min}) of the embryonic bubble inside the wall cavity. The size of the embryonic bubble is determined by the cone angle of the cavity (ϕ), the wettability of the liquid ammonia, and the size of the cavity mouth (R_c). Lorenz [26] developed a model to predict the initial radius of vapor trapped in the cavity by using the geometric parameters and contact angle of the liquid. In case the contact angle of the liquid (θ) is larger than the cone angle of the cavity, ϕ , the vapor can be trapped inside the cavity. He considered that the vapor is trapped when the advancing liquid front meets the opposite side of the cavity as depicted in Fig. 12 (a). As the volume trapped by the straight line is converted to the cone shape having an equivalent volume, the initial radius of the cone curvature is expressed in the following Eqs. [26].

$$\frac{r_{min}}{R_c} = f(\theta, \phi) = \left(g_1 / g_2 \right)^{1/3} \quad (8)$$

$$g_1(\theta, \phi) = \frac{\pi \sin(\theta - \phi) \tan^3 \psi (\tan \psi \cot \phi - 1)^2}{3 \sin(\phi/2) \cos \gamma (\tan^2 \phi - \tan^2 \gamma)^{3/2}} \quad (9)$$

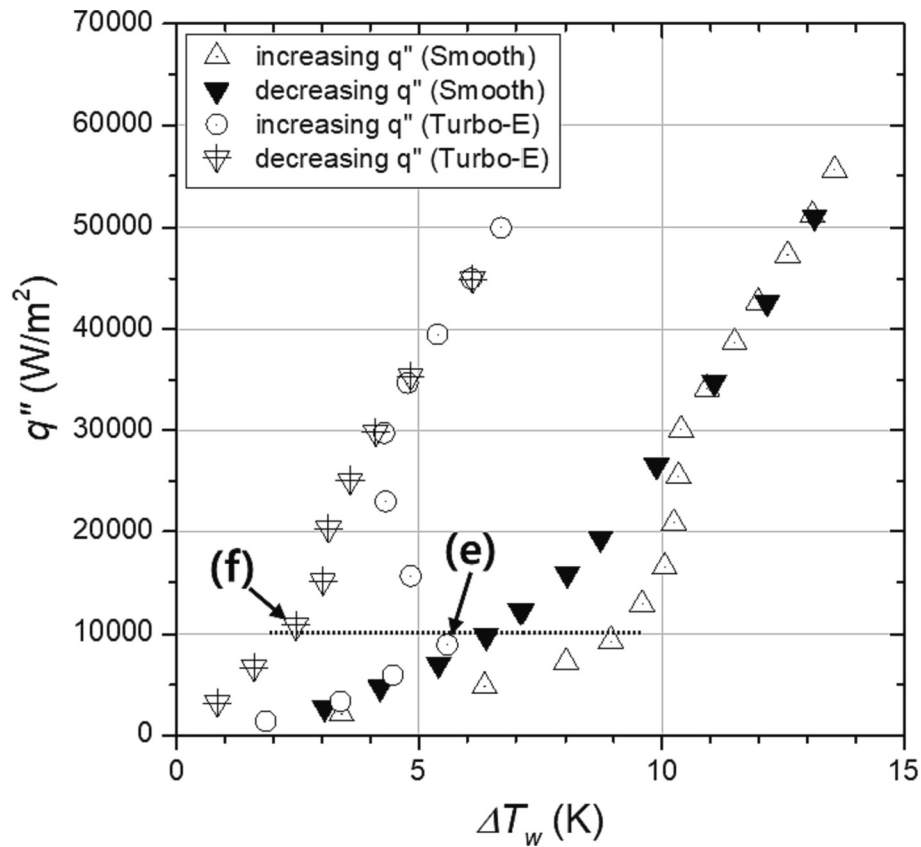


Fig. 9. Boiling curves on the Turbo-E tube compared with the smooth tube at the saturation pressure of 8.57 bar ($T_{sat} = 20^\circ\text{C}$).

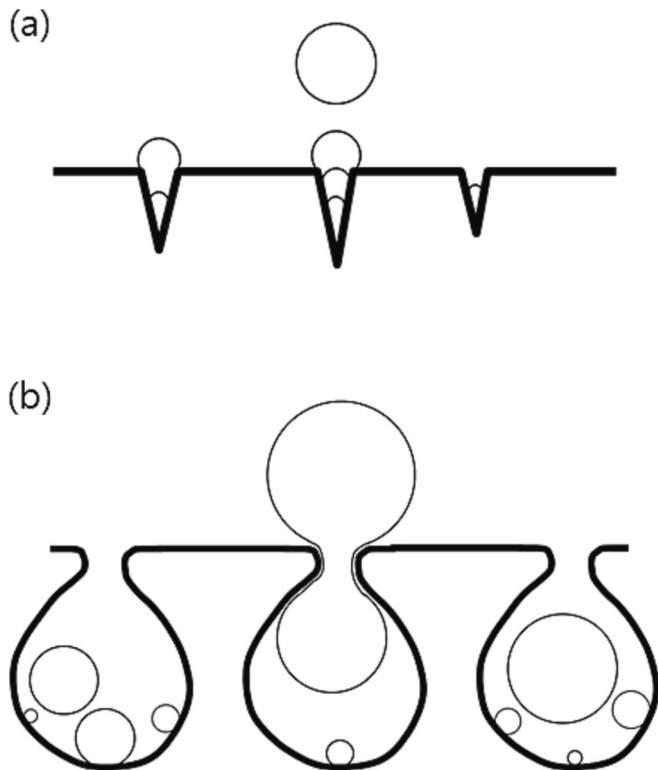


Fig. 10. Comparison of bubble nucleation process between (a) smooth tube or Low-fin tube and (b) Turbo-E tube.

$$\text{where } \varphi = 90 - \frac{\phi}{2}, \gamma = 90 - (\theta - \frac{\phi}{2})$$

$$g_2(\theta, \phi) = \frac{\pi}{3} \left[\frac{\cos^3(\theta - \phi/2)}{\tan(\phi/2)} + S^2(3 - S) \right] \quad (10)$$

$$\text{where } S = 1 - \sin(\theta - \frac{\phi}{2})$$

The contact angle of liquid ammonia is required to obtain the minimum radius of the embryonic bubble from Eq. (8). Despite the wide utilization of ammonia in the chemical plant and agricultural industry for a long time, measurement data of the equilibrium contact angle of ammonia on the stainless steel surface is rarely found. The main difficulty in obtaining the contact angle of ammonia seems to come from the toxicity and high pressure of ammonia. Fortunately, Ruzaikin et al. [27] reported the measurement of the equilibrium contact angle of ammonia on the stainless steel surface using the capillary rise method. They measured the contact angles of ammonia at three saturation temperatures of -11.3 , 54 , and 75.4°C . From the interpolation of their data, contact angles at three saturation temperatures of 10 , 20 , and 30°C are approximated to be 21.2° , 20.3° , and 19.4° , respectively. In case the cone angle of the cavity (ϕ) is 15° (so that $\phi < \theta < 90^\circ$), r_{min}/R_c at the contact angle ranging from 15° to 44° are displayed in Fig. 12(b). At the contact angle of 20.3° corresponding to the saturation temperature of 20°C , r_{min}/R_c is approximately 0.5 , which implies that the initial radius of the embryonic bubble equals half of the cavity radius. Then, the required wall superheat at the minimum radius ($r_{min} = 0.5R_c$) is calculated to be 10.7 K , which is consistent with the ΔT_w at the saturation temperature of 20°C acquired in section 3.1. As depicted in Fig. 12(b), if the contact angle of liquid ammonia on the solid surface is increased, the required wall superheat can be reduced. Polymers having low surface energy can be coated on the stainless steel tube to decrease the surface energy of the solid surface. For example, PTFE (Polytetrafluoroethylene) is widely used for making nonstick surfaces owing to its low surface

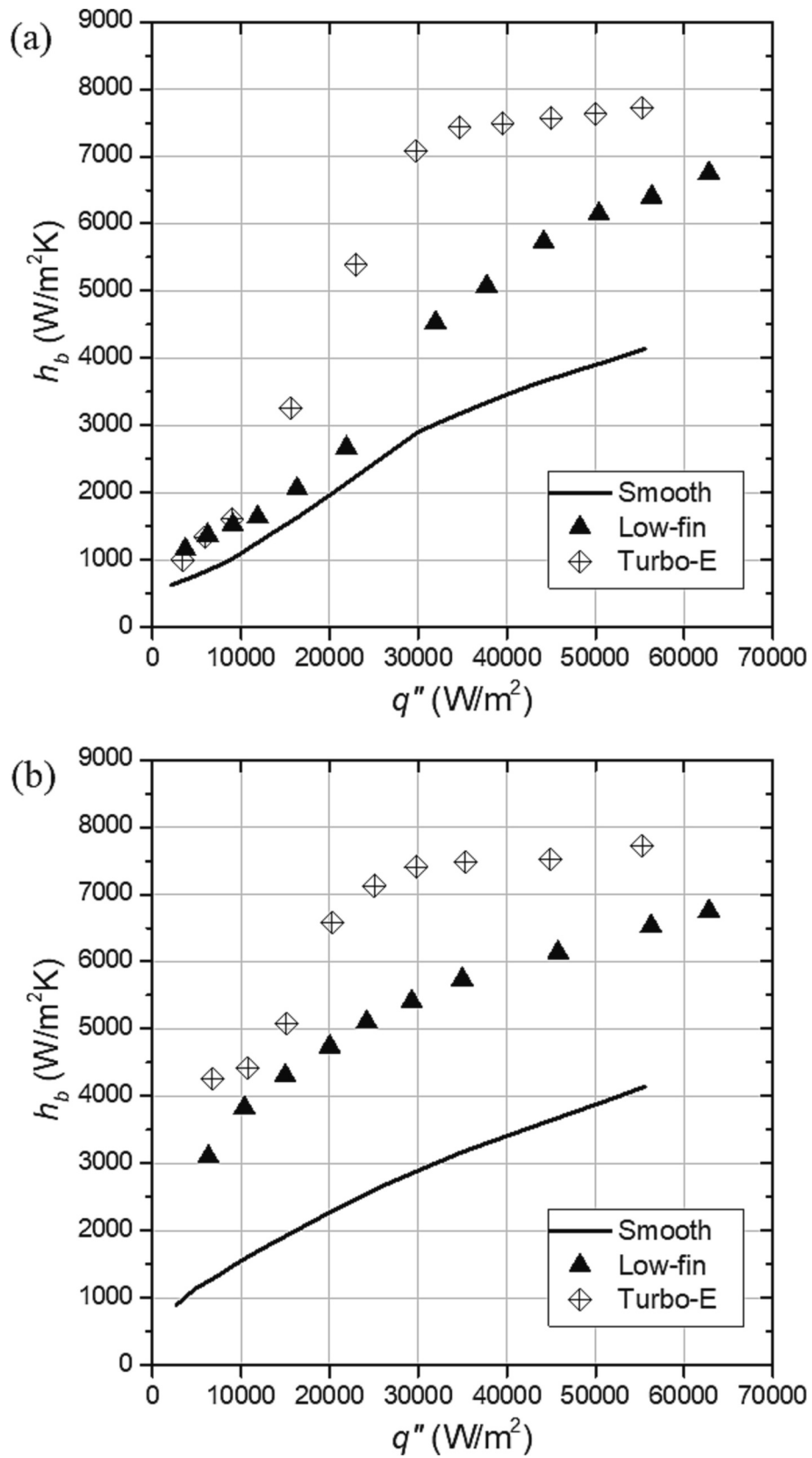


Fig. 11. Boiling heat transfer coefficients on three tubes (smooth, Low-fin, and Turbo-E) at the saturation pressure of 8.57 bar ($T_{sat} = 20$ °C), (a) increasing q'' and (b) decreasing q'' .

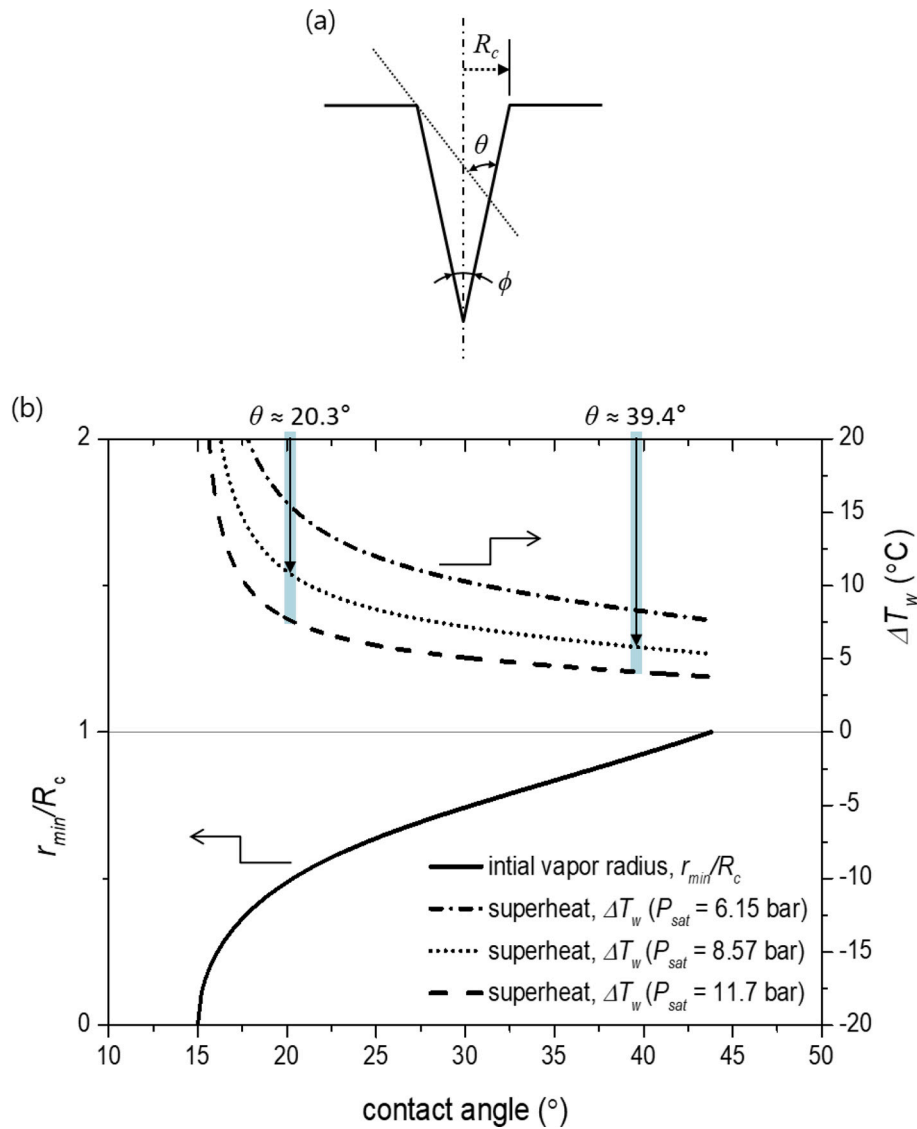


Fig. 12. (a) Schematic drawing of the initial vapor trapping process in the cavity, (b) minimum bubble radius and wall superheat according to the contact angle (θ) at the cavity cone angle (ϕ) of 15° .

energy of approximately 18 mN/m. The surface energy of PTFE is slightly lower than that of liquid ammonia ($\sigma = 21.7$ mN/m at $T_{sat} = 20^\circ\text{C}$). Although the contact angle of ammonia on the PTFE surface is not known, Carrera et al. [28] reported contact angles on the PTFE surface lay between 39.7° and 80.5° for various ionic liquids with surface energy ranging from 21.4 to 46.1 mN/m. From the quadratic curve fitting of the measured contact angle data [29], the contact angle of ammonia having a surface energy of 21.7 mN/m is predicted to be 39.4° . As marked in Fig. 12(b), the required wall superheat decreases down to 5.8 K at the calculated contact angle on the PTFE-coated surface, which is almost half of the wall superheat on the bare stainless steel tube (10.7 K). Therefore, it is expected that the bubble nucleation is initiated at the lower wall temperature in case of increasing the heat flux, resulting in the alleviation of the hysteresis in a boiling curve.

Another way to mitigate the hysteresis in a boiling curve of ammonia is to employ the re-entrant cavity with a small cavity mouth. The size of the cavity mouth is as high as 100–150 μm for the present Turbo-E tube. While the Turbo-E tube has numerous re-entrant cavities on the wall, the cavity mouth is too large to trap vapor effectively from highly wetting liquids such as ammonia. Since the present Turbo-E tube is optimized for an evaporator using water, the geometric parameters of the re-entrant cavity should be optimized to minimize the hysteresis in ammonia. If

the size of the cavity mouth is reduced to a few micrometers, a larger vapor bubble can be trapped inside the re-entrant cavity, even when employed in ammonia. In addition, it is expected that the Turbo-E tube with smaller cavity mouths improves the boiling heat transfer performance in ammonia. Considering the difficulty in machining re-entrant cavities having such a small mouth of a few micrometers, porous coating using microscale metallic powders can be a viable measure to make re-entrant cavities on the tube.

4. Conclusions

Pool boiling heat transfer of ammonia outside a single tube was investigated on both a smooth tube and two enhanced tubes (Low-fin and Turbo-E tubes). The following conclusions are deduced from the test results.

- Large hysteresis between when the heat flux increases and when the heat flux decreases is observed in a boiling curve of ammonia on the smooth tube. A large superheat is required when increasing the heat flux because a bubble grows from the small embryonic bubble, while a smaller superheat is required when decreasing the heat flux

because a bubble grows from the large remaining bubble after the preceding departure.

- The hysteresis in a boiling curve of ammonia is reduced as the saturation pressure increases. Maximum wall superheat when increasing the heat flux decreases from 13.9 K ($P_{sat} = 6.15$ bar) to 8.9 K ($P_{sat} = 11.7$ bar), which is comparable to the required wall superheats obtained from the thermodynamic analysis.
- Heat transfer coefficients are augmented by 47% and 103% (increasing q'') and 105% and 159% (decreasing q''), respectively, for two enhanced tubes, Low-fin and Turbo-E compared with the smooth tube. On the contrary, the hysteresis in a boiling curve is increased on the enhanced tubes owing to the wall temperature decrease along the fin.
- A hydrophobic coating like a PTFE layer and the enhanced tube having re-entrant cavities with small cavity mouths of a few micrometers are suggested to mitigate the hysteresis in a boiling curve of ammonia.

CRediT authorship contribution statement

Jin Sub Kim: Conceptualization, Methodology, Investigation, Writing – original draft. **Wookyoung Kim:** Visualization, Investigation. **Hak Soo Kim:** Software, Validation. **Young Kim:** Writing – review & editing. **Seok Ho Yoon:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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